

WASP-12b

R-1286

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_200126 · created 2026-07-31T15:41:51+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458874.65533 +/- 0.0007 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.044
Transit depth (Rp/Rs)^2	5.2e-05 +/- 0.0064 [%]
Orbital Inclination (inc)	80.61 +/- 2.89
Airmass coefficient 1 (a1)	1.92 +/- 1.19
Airmass coefficient 2 (a2)	-0.69 +/- 0.72
Scatter in the residuals of the lightcurve fit is	55.84 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.105 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.044 vs 0.1178 (deviation 2.65σ)
Duration fit vs published	0.105 d vs 0.125 d (deviation 1.82σ)
Mid-time O–C	0.0 min
Depth SNR	0.0
Residual scatter	55.84 %

Light curve

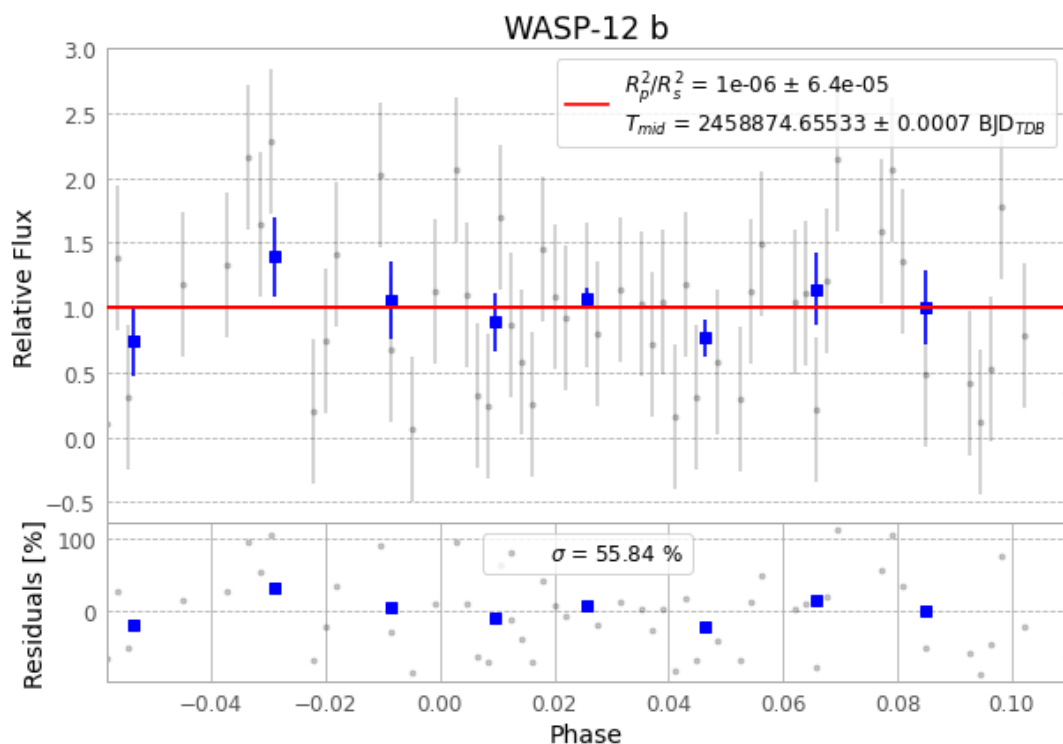


Figure 1 — FinalLightCurve_WASP-12 b_2020-01-25.png (SHA-256 b803c412548d1a32...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 303], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dfa847a378d68810...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), WASP-12200126015443.FITS → WASP-12200126062712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-200126.log (f989845801a8...), FinalLightCurve_WASP-12 b_2020-01-25.png (b803c412548d...), FinalLightCurve_WASP-12 b_2020-01-25.csv (a0396aae8a05...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 15:41Z.